where y_{s_1} is the normalized input admittance of the shunt element. To match the feed line to the parallel combination, we need $y_{in_1} = 1 + j0$. Hence, we need y_{s_1} to cancel the imaginary part of y_{d_1}; that is,

$$y_{s_1} = -j1.58.$$

The corresponding impedance of the lumped element is

$$Z_{s_1} = \frac{1}{Y_{s_1}} = \frac{1}{y_{s_1} Y_0} = \frac{Z_0}{jb_{s_1}} = \frac{Z_0}{-j1.58} = \frac{jZ_0}{1.58} = j31.62\ \Omega.$$

Since the value of Z_{s_1} is positive, the element to be inserted should be an inductor and its value should be

$$L = \frac{31.62}{\omega} = \frac{31.62}{2\pi \times 10^8} = 50\ \text{nH}.$$

The results of this solution have been incorporated into the circuit of **Fig. 2-35**.

Solution for Point *D* (Fig. 2-36): At point *D*,

$$y_{d_2} = 1 - j1.58,$$

and the distance between points *B* and *D* is

$$d_2 = (0.322 - 0.115)\lambda = 0.207\lambda.$$

The needed normalized admittance of the reactive element is

$$y_{s_2} = +j1.58.$$

Hence,

$$Z_{s_2} = -j31.62\ \Omega,$$

which is the impedance of a capacitor with

$$C = \frac{1}{31.62\omega} = 50\ \text{pF}.$$

Figure 2-36 displays the circuit solution for d_2 and C.

2-11.2 Single-Stub Matching

The single-stub matching network shown in **Fig. 2-37(a)** consists of two transmission line sections: one of length d connecting the load to the feedline at MM' and another of length l connected in parallel with the other two lines at MM'. This second line is called a ***stub***, and it is usually terminated in either a short or open circuit; hence, its input impedance and admittance are purely reactive. The stub shown in **Fig. 2-37(a)** has a short-circuit termination.

► The required two degrees of freedom are provided by the length l of the stub and the distance d from the load to the stub position. ◄

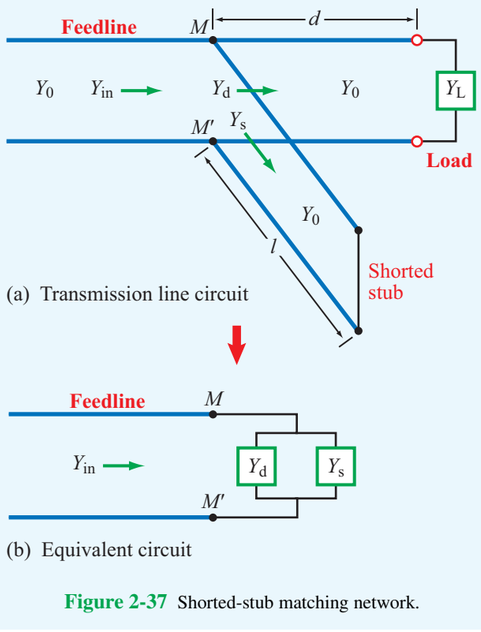


Figure 2-37 Shorted-stub matching network.

Because at MM' the stub is added in parallel to the line (which is why it is called a ***shunt*** stub), it is easier to work with admittances than with impedances. The matching procedure consists of two steps. In the first step, the distance d is selected to transform the load admittance, $Y_L = 1/Z_L$, into an admittance of the form $Y_d = Y_0 + jB$ when looking toward the load at MM'. Then in the second step, the length l of the stub line is selected so that its input admittance Y_s at MM' is equal to $-jB$. The parallel sum of the two admittances at MM' yields Y_0, which is the characteristic admittance of the line. The procedure is illustrated by Example 2-14.

Example 2-14: Single-Stub Matching

Repeat Example 2-13, but use a shorted stub (instead of a lumped element) to match the load impedance $Z_L = (25 - j50)\ \Omega$ to the 50 Ω transmission line.

Solution: In Example 2-13, we demonstrated that the load can be matched to the line via either of two solutions:

1. $d_1 = 0.063\lambda$, and $y_{s_1} = jb_{s_1} = -j1.58$,

其中 y_{s_1} 是分流元件的归一化输入导纳。为了将馈线与并联组合相匹配，我们需要 $y_{in_1} = 1 + j0$。因此，我们需要 y_{s_1}来抵消y_{d_1}的虚部；那是，

$$y_{s_1} = -j1.58.$$

集总元件的相应阻抗为

$$Z_{s_1} = \frac{1}{Y_{s_1}} = \frac{1}{y_{s_1} Y_0} = \frac{Z_0}{jb_{s_1}} = \frac{Z_0}{-j1.58} = \frac{jZ_0}{1.58} = j31.62\ \Omega.$$

由于Z_{s_1}的值为正，因此要插入的元件应该是电感器，其值应该是

$$L = \frac{31.62}{\omega} = \frac{31.62}{2\pi \times 10^8} = 50\ \text{nH}.$$

该解决方案的结果已合并到图 2-35 的电路中。

D 点的解（图 2-36）： 在 D 点，

$$y_{d_2} = 1 - j1.58,$$

B 点和 D 点之间的距离为

$$d_2 = (0.322 - 0.115)\lambda = 0.207\lambda.$$

电抗元件所需的归一化导纳为

$$y_{s_2} = +j1.58.$$

因此，

$$Z_{s_2} = -j31.62\ \Omega,$$

这是电容器的阻抗

$$C = \frac{1}{31.62\omega} = 50\ \text{pF}.$$

图 2-36 显示了 d_2 和 C 的电路解决方案。

2-11.2 单存根匹配

图2-37（a）所示的单短截线匹配网络由两个传输线部分组成：一个长度为d，在MM′处将负载连接到馈线，另一个长度为l，在MM′处与其他两条线路并联连接。第二条线称为短线，通常以短路或开路终止；因此，其输入阻抗和导纳是纯电抗性的。图2-37(a)所示的短截线具有短路终端。

► 所需的两个自由度由短截线的长度 l 和从负载到短截线位置的距离 d 提供。 ◄

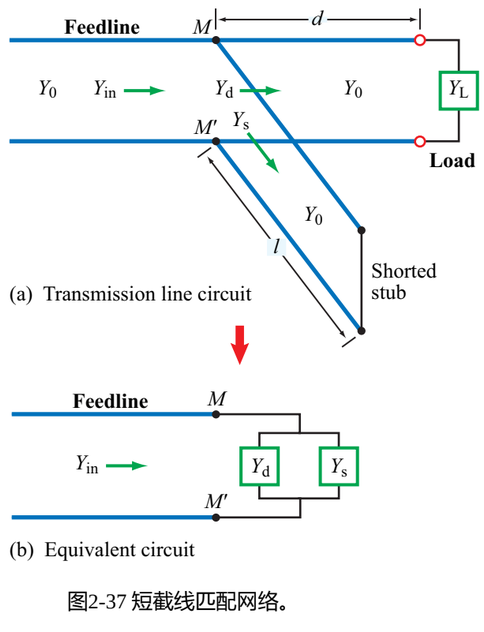


图2-37 短截线匹配网络。

因为在 MM′ 处，短截线与线路并联添加（这就是它被称为并联短截线的原因），因此使用导纳比使用阻抗更容易。匹配过程由两个步骤组成。第一步，选择距离 d，将负载导纳 $Y_L = 1/Z_L$ 转换为 $Y_d = Y_0 + jB$ 形式的导纳（当观察 MM′ 处的负载时）。然后在第二步中，选择短截线的长度l，使其在MM′处的输入导纳Y_s等于−jB。 MM′ 处两个导纳的并联和得出 Y_0，它是线路的特征导纳。示例 2-14 说明了该过程。

示例 2-14：单存根匹配

重复示例 2-13，但使用短路短线（而不是集总元件）将负载阻抗 $Z_L = (25 - j50)\ \Omega$ 与 50 Ω 传输线相匹配。

解决方案：在示例 2-13 中，我们演示了可以通过以下两种解决方案之一将负载与线路匹配：

1. $d_1 = 0.063\lambda$, and $y_{s_1} = jb_{s_1} = -j1.58$,

2. $d_2 = 0.207\lambda$, and $y_{s_2} = jb_{s_2} = j1.58$.

The locations of the insertion points at distances d_1 and d_2 remain the same, but now our task is to select corresponding lengths l_1 and l_2 of shorted stubs that present the required admittances at their inputs.

To determine l_1, we use the Smith chart in **Fig. 2-38**. The normalized admittance of a short circuit is $-j\infty$, which is represented by point E on the Smith chart with position 0.25λ on the WTG scale. A normalized input admittance of $-j1.58$ is located at point F with position 0.34λ on the WTG scale. Hence,

$$l_1 = (0.34 - 0.25)\lambda = 0.09\lambda.$$

Similarly, $y_{s_2} = j1.58$ is represented by point G with position 0.16λ on the WTG scale of the Smith chart in **Fig. 2-39**. Rotating from point E to point G involves a rotation of 0.25λ plus an additional rotation of 0.16λ or

$$l_2 = (0.25 + 0.16)\lambda = 0.41\lambda.$$

Example 2-15: $\lambda/4$ **Transformer for Complex Load**

Design a quarter-wavelength transformer to match a load with $Z_L = (100 + j100)\ \Omega$ to a 50 Ω line.

Solution: The required transmission-line circuit is shown in **Fig. 2-40(a)** on p. 105. We need to insert a $\lambda/4$ transformer to eliminate reflections at MM'. To that end, we need to specify the insertion distance d and the characteristic impedance Z_{0_2} of the $\lambda/4$ section so that $Z_{in} = Z_{0_1}$.

A $\lambda/4$ section with characteristic impedance Z_{0_2} transfers an impedance Z_a at one end of the section to impedance Z_b at the other end such that

$$Z_a Z_b = Z_{0_2}^2. \tag{2.142}$$

In the present case (**Fig. 2-40(a)**), $Z_a = Z(d)$, the input impedance at BB' looking towards the load, and $Z_b = Z_{in} = Z_{0_1} = 50\ \Omega$. Since the $\lambda/4$ transformer is a lossless line, its impedance Z_{0_2} is purely real. Consequently, to satisfy Eq. (2.142), it is necessary that $Z(d)$ be purely real as well, which is the key to how we select the distance d. Given this rationale, we proceed as follows.

1. The normalized load impedance is

$$z_L = \frac{Z_L}{Z_{0_1}} = \frac{100 + j100}{50} = 2 + j2,$$

which is shown as point A in the Smith chart of **Fig. 2-40(b)**.

2. Move on the SWR circle until $z(d)$ becomes purely real, which occurs at points B and C.

(a) Point *B*:

$$\begin{aligned} d_1 &= (0.25 - 0.209)\lambda \\ &= 0.041\lambda, \\ z(d_1) &= 4.27, \\ Z(d_1) &= 4.27 \times 50 = 213.5\ \Omega, \\ Z_{0_2} &= \sqrt{Z(d_1)\, Z_{0_1}} \\ &= \sqrt{213.5 \times 50} \\ &= 103.3\ \Omega. \end{aligned}$$

(b) Point *C*:

$$\begin{aligned} d_2 &= d_1 + \frac{\lambda}{4} \\ &= (0.041 + 0.25)\lambda = 0.291\lambda, \\ z(d_2) &= 0.23, \\ Z(d_2) &= 0.23 \times 50 = 11.5\ \Omega, \\ Z_{0_2} &= \sqrt{Z(d_2)\, Z_{0_1}} \\ &= \sqrt{11.5 \times 50} \\ &= 24.2\ \Omega. \end{aligned}$$

Concept Question 2-24: To match an arbitrary load impedance to a lossless transmission line through a matching network, what is the required minimum number of degrees of freedom that the network should provide?

Concept Question 2-25: In the case of the single-stub matching network, what are the two degrees of freedom?

Concept Question 2-26: When a transmission line is matched to a load through a single-stub matching network, no waves are reflected toward the generator. What happens to the waves reflected by the load and by the shorted stub when they arrive at terminals MM' in **Fig. 2-37**?

102 CHAPTER 2 TRANSMISSION LINES

2. $d_2 = 0.207\lambda$, and $y_{s_2} = jb_{s_2} = j1.58$.

距离 d_1 和 d_2 处的插入点位置保持不变，但现在我们的任务是选择短路短截线的相应长度 l_1 和 l_2，以在其输入端提供所需的导纳。

为了确定 l_1，我们使用图 2-38 中的史密斯圆图。短路的归一化导纳为 $-j\infty$，它由史密斯圆图上 WTG 标度上位置 0.25λ 上的点 E 表示。归一化输入导纳 $-j1.58$ 位于 WTG 标度上位置 0.34λ 的 F 点。因此，

$$l_1 = (0.34 - 0.25)\lambda = 0.09\lambda.$$

类似地，$y_{s_2} = j1.58$ 由图 2-39 中史密斯圆图 WTG 标度上位置 0.16λ 的 G 点表示。从 E 点旋转到 G 点需要旋转 0.25λ 加上额外的 0.16λ 旋转，或者

$$l_2 = (0.25 + 0.16)\lambda = 0.41\lambda.$$

示例 2-15：复杂负载的 $\lambda/4$ 变压器

设计一个四分之一波长变压器，将 $Z_L = (100 + j100)\ \Omega$ 的负载与 50 Ω 线路相匹配。

解：所需的传输线电路如图2-40(a)所示。 105. 我们需要插入一个 $\lambda/4$ 变压器来消除 MM' 处的反射。为此，我们需要指定插入距离 d 和 $\lambda/4$ 部分的特性阻抗 Z_{0_2}，以便 $Z_{in} = Z_0$

具有特性阻抗 Z_{0_2} 的 $\lambda/4$ 部分[1]将该部分一端的阻抗 Z_a 传输到另一端的阻抗 Z_b，使得

$$Z_a Z_b = Z_{0_2}^2. \tag{2.142}$$

在本例中（图 2-40(a)），$Z_a = Z(d)$，BB' 处朝向负载的输入阻抗，以及 $Z_b = Z_{in} = Z_{0_1} = 50\ \Omega$。由于$\lambda/4$变压器是无损线路，因此其阻抗$Z_{0_2}$是纯实数。因此，满足方程 。 (2.142)，Z(d)也必须是纯实数，这是我们如何选择距离d的关键。考虑到这个理由，我们进行如下操作。

1. 归一化负载阻抗为

$$z_L = \frac{Z_L}{Z_{0_1}} = \frac{100 + j100}{50} = 2 + j2,$$

如图2-40(b)史密斯圆图中的A点所示。

2. 在 SWR 圆上移动，直到 z(d) 变为纯实数，这发生在 B 点和 C 点。

(a) B点：

$$\begin{aligned} d_1 &= (0.25 - 0.209)\lambda \\ &= 0.041\lambda, \\ z(d_1) &= 4.27, \\ Z(d_1) &= 4.27 \times 50 = 213.5\ \Omega, \\ Z_{0_2} &= \sqrt{Z(d_1)\, Z_{0_1}} \\ &= \sqrt{213.5 \times 50} \\ &= 103.3\ \Omega. \end{aligned}$$

(b) C点：

$$\begin{aligned} d_2 &= d_1 + \frac{\lambda}{4} \\ &= (0.041 + 0.25)\lambda = 0.291\lambda, \\ z(d_2) &= 0.23, \\ Z(d_2) &= 0.23 \times 50 = 11.5\ \Omega, \\ Z_{0_2} &= \sqrt{Z(d_2)\, Z_{0_1}} \\ &= \sqrt{11.5 \times 50} \\ &= 24.2\ \Omega. \end{aligned}$$

概念问题 2-24：为了通过匹配网络将任意负载阻抗与无损传输线相匹配，网络应提供的最小自由度数是多少？

概念问题 2-25：在单根匹配网络的情况下，两个自由度是多少？

概念问题 2-26：当传输线通过单短截线匹配网络与负载匹配时，不会有波反射到发电机。当负载和短路短截线反射的波到达图 2-37 中的端子 MM' 时，会发生什么情况？

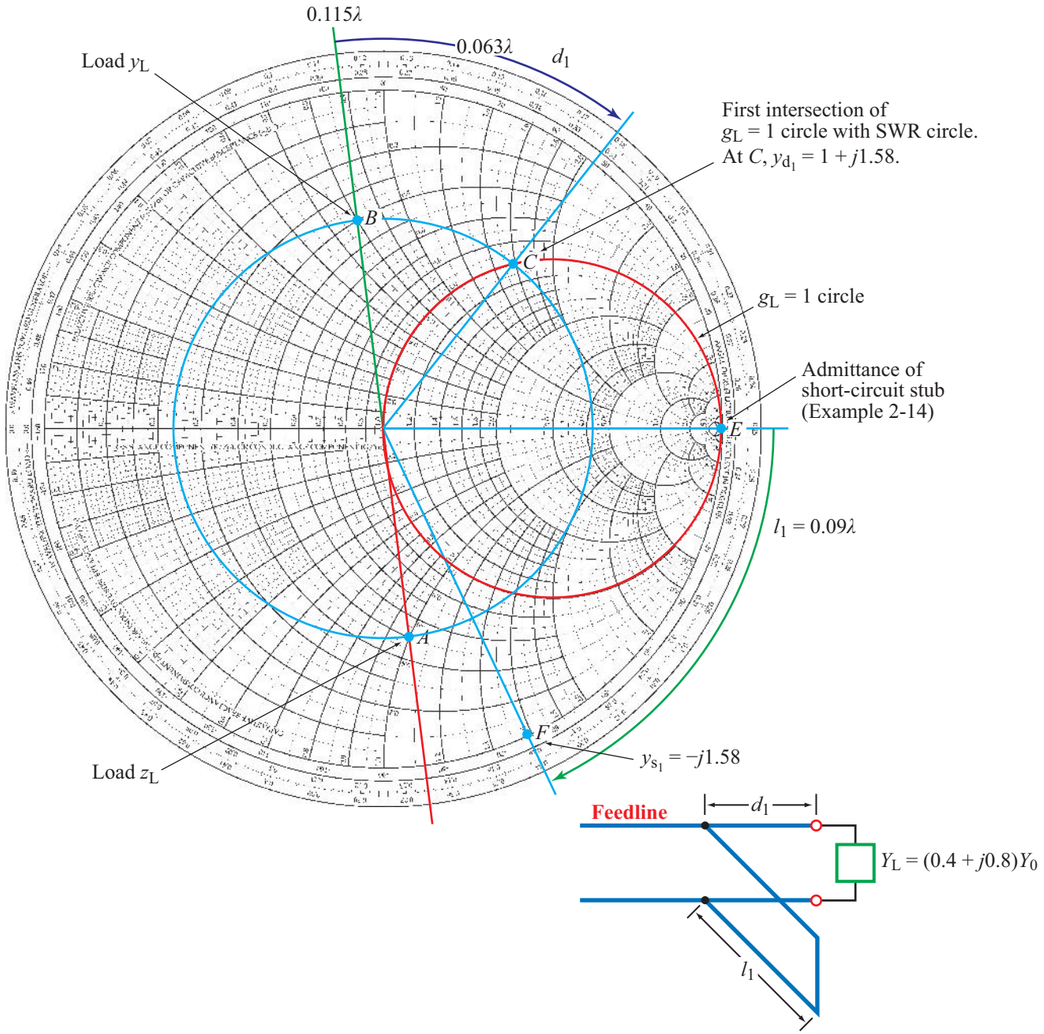


Figure 2-38 Solution for point C of Example 2-14. Point A is the normalized load with $z_L = 0.5 - j1$; point B is $y_L = 0.4 + j0.8$. Point C is the intersection of the SWR circle with the $g_L = 1$ circle. The distance from B to C is $d_1 = 0.063\lambda$. The length of the shorted stub (E to F) is $l_1 = 0.09\lambda$.

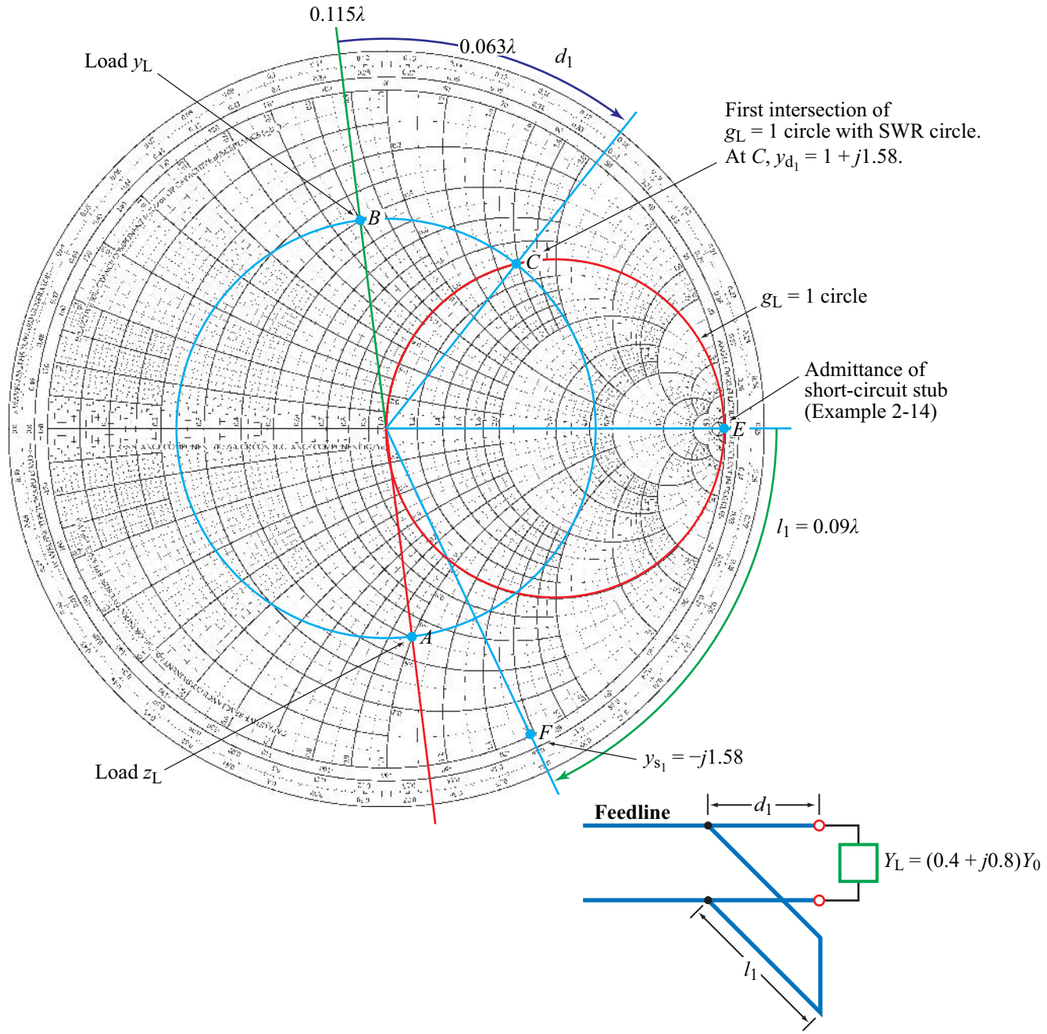


图 2-38 例 2-14 中 C 点的解法。 A点为$z_L = 0.5 - j1$的归一化载荷； B 点为 $y_L = 0.4 + j0.8$。 C 点是 SWR 圆与 $g_L = 1$ 圆的交点。 B 到 C 的距离为 $d_1 = 0.063\lambda$。短路短截线（E 至 F）的长度为 $l_1 = 0.09\lambda$。

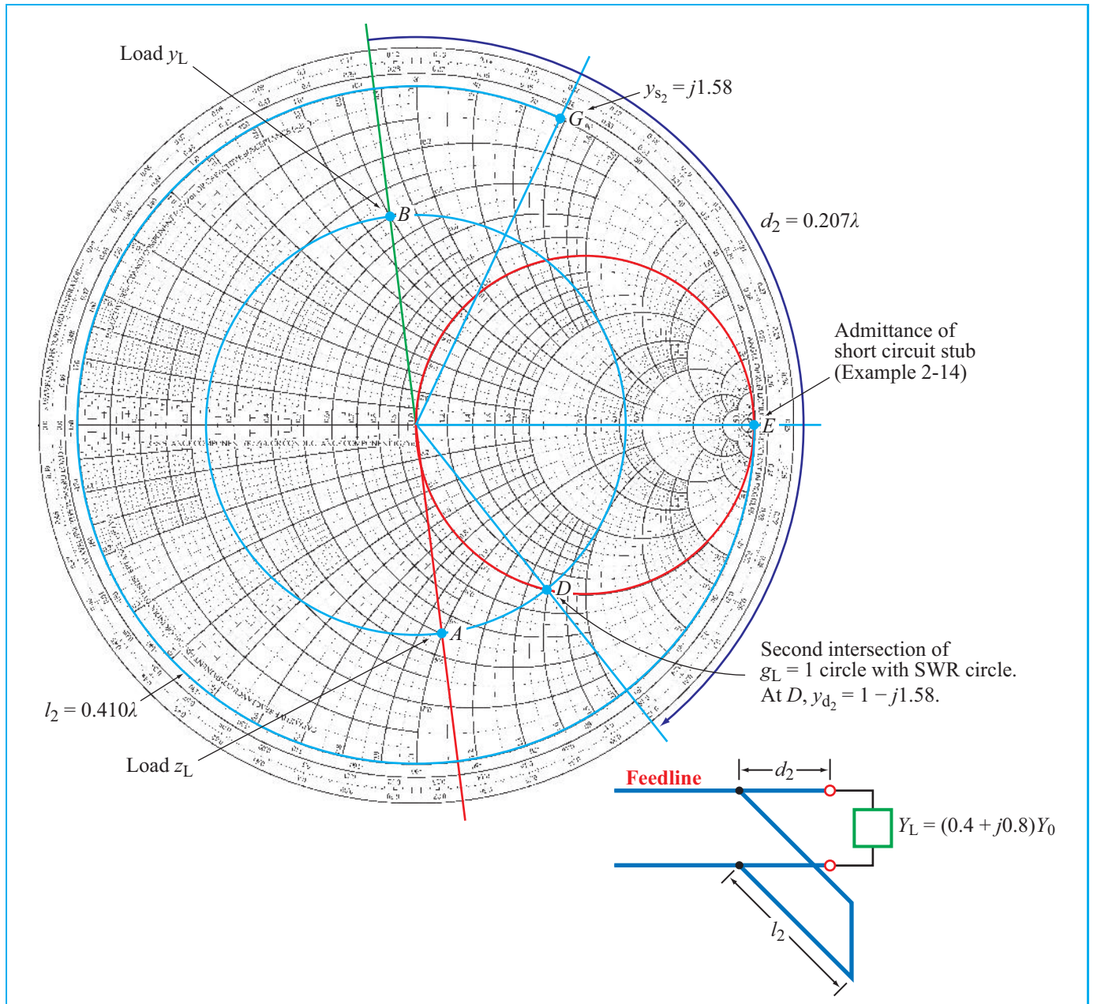


Figure 2-39 Solution for point *D* of Example 2-14. Point *D* is the second point of intersection of the SWR circle and the $g_L = 1$ circle. The distance *B* to *D* gives $d_2 = 0.207\lambda$, and the distance *E* to *G* gives $l_2 = 0.410\lambda$.

104 CHAPTER 2 TRANSMISSION LINES

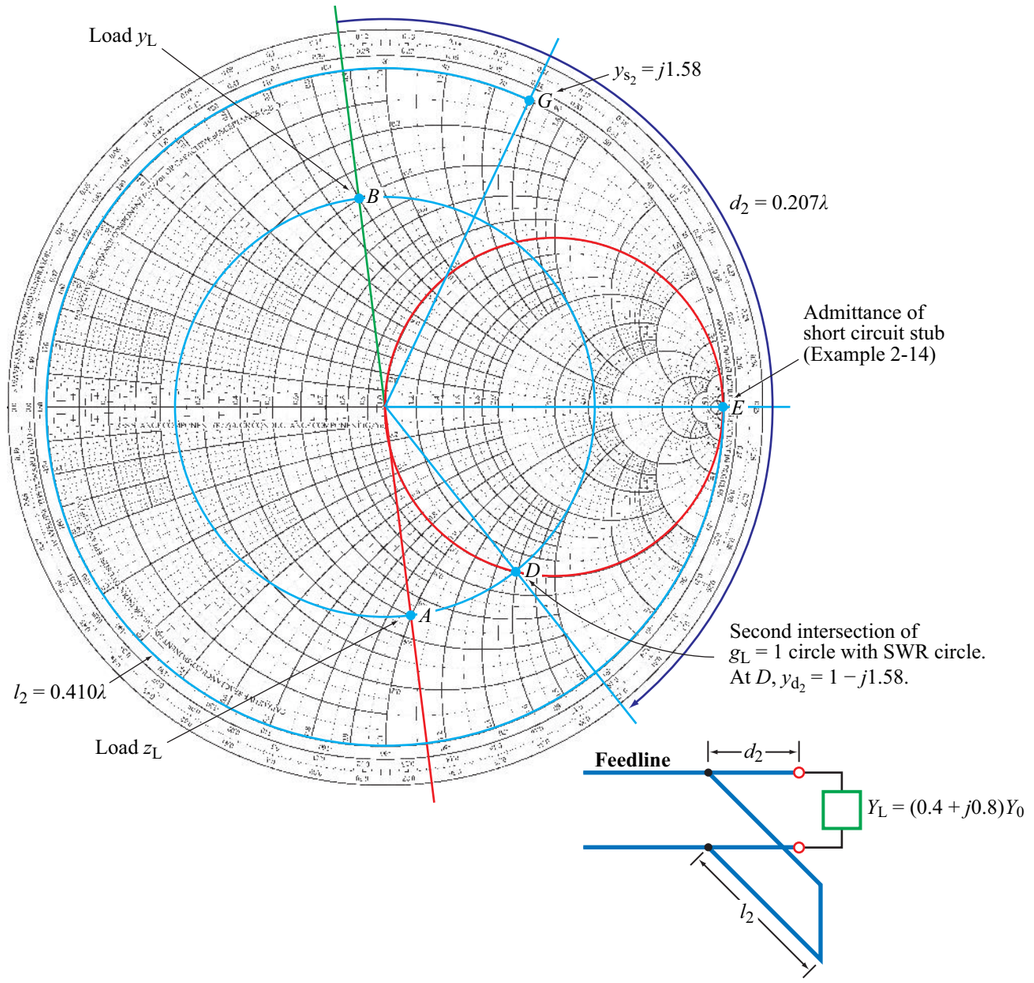
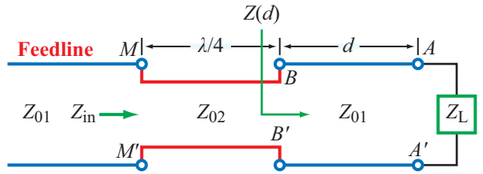
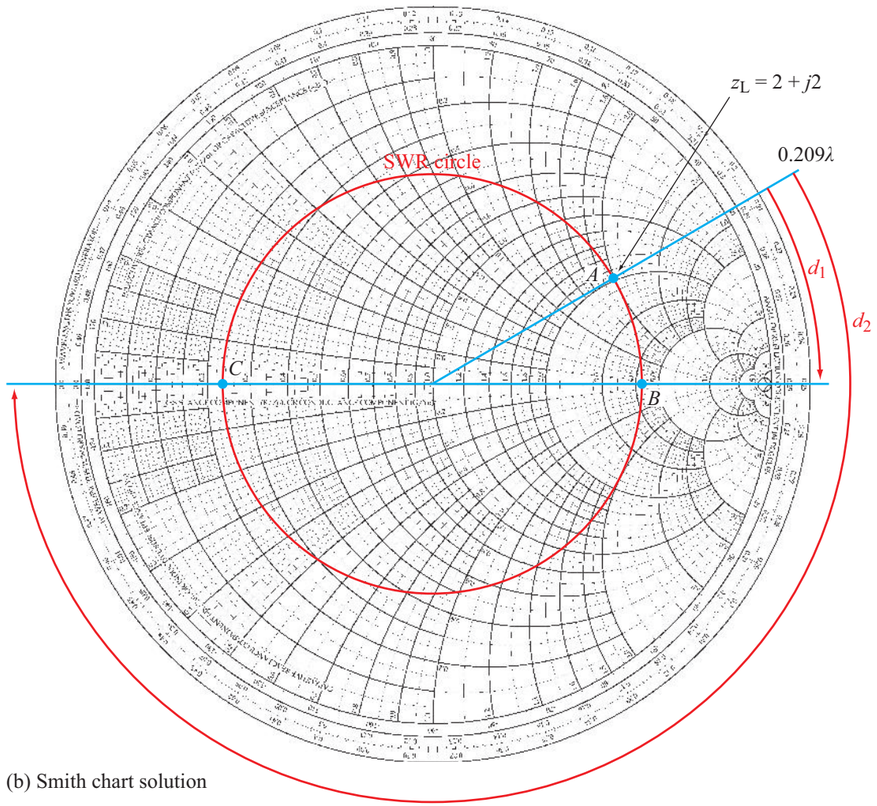


图 2-39 例 2-14 中 D 点的解法。点D是SWR圆和$g_L = 1$圆的第二个交点。 B 到 D 的距离为 $d_2 = 0.207\lambda$， E 到 G 的距离为 $l_2 = 0.410\lambda$。

(a) Quarter-wave transformer

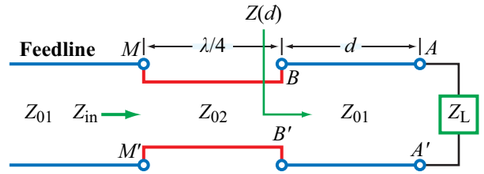


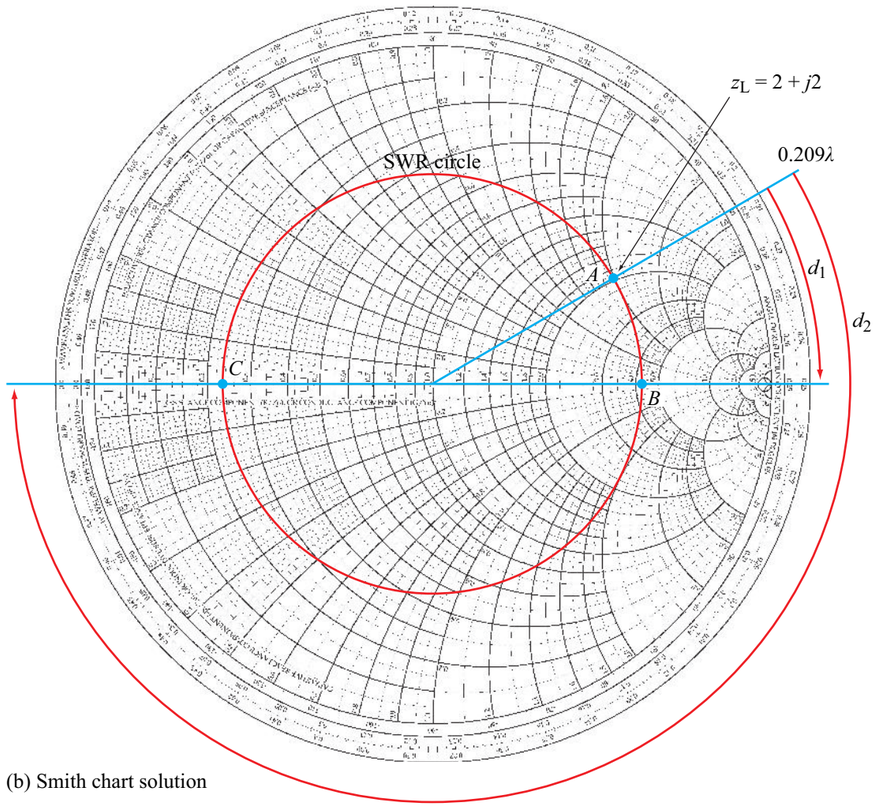


(b) Smith chart solution

Figure 2-40 Solution for Example 2-15: (a) quarter-wave transformer and (b) Smith chart solution.

(a) Quarter-wave transformer





(b) Smith chart solution

图 2-40 示例 2-15 的解决方案：(a) 四分之一波长变压器和 (b) 史密斯圆图解决方案。